Iterating semi-discrete JKO steps by stochastic approximation

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November 12th

(Euclidean) Gradient flows

$$E : \mathbb{R}^d \to \mathbb{R}, \qquad \dot{x}_t = -\nabla E(x_t)$$

- **Explicit Euler:** $\tau > 0$, $\nabla E(x_n^\tau) + \frac{x_{n+1}^\tau - x_n^\tau}{\tau} = 0$.
  Needs E to be differentiable.

- **Implicit Euler:** $\tau > 0$, $\nabla E(x_{n+1}^\tau) + \frac{x_{n+1}^\tau - x_n^\tau}{\tau} = 0$

- Equivalent to the variational problem

$$x_{n+1}^\tau = \operatorname*{arg\,min}_{x \in \mathbb{R}^d} E(x) + \frac{1}{2\tau}\|x - x_n^\tau\|^2$$

  Only needs a distance on $\mathbb{R}^d$!

(Wasserstein) Gradient flows

We define recursively, starting from $\mu \in \mathcal{P}_2(\mathbb{R}^d)$

$$\rho_0^\tau = \mu, \qquad \rho_{n+1}^\tau = \operatorname*{arg\,min}_{\rho \in \mathcal{P}_2(\mathbb{R}^d)} \mathcal{E}(\rho) + \frac{1}{2\tau} W_2(\rho, \rho_n^\tau)^2$$

- $W_2(\mu, \nu)$ is the optimal transport distance.

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- Piecewise constant interpolation of the $(\rho_n^\tau)_n$ and letting $\tau \to 0$ yields a curve in $\mathcal{P}_2(\mathbb{R}^d)$ solving (weakly) the continuity equation

  $$\partial_t \rho_t = -\mathrm{div}(\rho_t v_t)$$

  for some vector field v_t (related to $\mathcal{E}$).

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  for some vector field v_t (related to $\mathcal{E}$).
- $\mathcal{E}(\rho) = \int V\rho + \int \rho \ln(\rho)$ yields the Fokker–Planck equation

  $$\partial_t \rho_t = \Delta \rho_t + \mathrm{div}(x \rho_t)$$

One (general) JKO step 1/2

- Recall, given a time-step $\tau > 0$, we are interested in solving

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg\min_\rho \ \mathcal{H}(\rho|\gamma) + \dfrac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

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- **At step n:** We show that

$$\rho_{n+1}^\tau \propto \exp\left(-(\varphi^\star)^{c/2\tau}\right) \mathrm{d}\gamma$$

where $\varphi^\star$ is a Kantorovich potential satisfying

$$\varphi^\star \in \arg\min_\varphi \ \tilde{J}(\varphi) := \ln\left(\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \varphi \mathrm{d}\rho_n^\tau.$$

One (general) JKO step 2/2

Proposition ***(G. 26+)***

The functional $\tilde{J}(\varphi) := \ln\left(\int e^{-\varphi^{c/2\tau}}\mathrm{d}\gamma\right) - \int \varphi \mathrm{d}\rho_n^\tau$ is convex.

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Theorem *(G. 26+)*

Let $J(\varphi) := \int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma - \int \varphi \mathrm{d}\rho_n^\tau$. Then,

$$\arg\min_\varphi J(\varphi) \subsetneq \arg\min_\varphi \tilde{J}(\varphi) \quad \text{and} \quad \arg\min_{\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma = 1} \tilde{J}(\varphi) = \arg\min_\varphi J(\varphi).$$

Pro: J is "SGDisable"

Con: one more parameter to estimate (the normalisation constant!)

One semi-discrete JKO step 1/3

- $J(\varphi) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(-\varphi^{c/2\tau}(y)\right) - \int \varphi(x)\rho_n^\tau(\mathrm{d}x)\right]$

One semi-discrete JKO step 1/3

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- Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n = \frac{1}{N}\sum_{i=1}^{N}\delta_{x_i} \rightsquigarrow \begin{cases} \varphi(x_i) = \theta_i \\ \varphi^{c/2\tau}(y) = \min_i\left\{\frac{c(x_i,y)}{2\tau} - \theta_i\right\} \end{cases}$$

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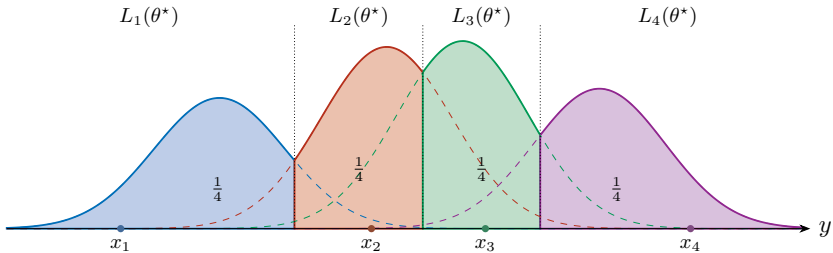
- $G(\theta) := J(\varphi(\cdot\,;\theta)) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(\max_{1\le i\le N}\left\{\theta_i - \frac{c(x_i,y)}{2\tau}\right\}\right) - \frac{1}{N}\sum_{i=1}^N \theta_i\right]$

One semi-discrete JKO step 2/3

- The next JKO step is

$$\rho^\star(y) \propto e^{\max_i \theta_i^\star - \frac{c(x_i,y)}{2\tau}} \gamma(y),$$

where $\theta^\star$ minimises G.



One semi-discrete JKO step 3/3

- The minimiser $\theta^\star$ is such that

$$\forall i, \qquad \rho^\star(L_i(\theta^\star)) = \frac{1}{N}.$$

- $\theta^\star \in \mathcal{B}$ which is explicit

